

Week 3: Introduction to Data Analysis

Sales Data Analysis

1. Project Overview

This project was developed as part of Week 3 of my Data Science Internship at The Developers Arena.

The project focuses on analyzing a real sales dataset using Python and Pandas. The analysis includes loading the CSV file, exploring the dataset, checking data quality, calculating sales metrics, identifying the best-selling product, and generating useful business insights.

The project demonstrates basic data analysis concepts such as data exploration, data cleaning, descriptive statistics, aggregation, and extracting insights from structured data.

2. Objectives

The main objectives of this project are:

- To understand the basics of data analysis.
- To load and analyze CSV data using Pandas.
- To explore rows, columns, and data types.
- To check for missing values and duplicate records.
- To calculate useful sales metrics.
- To identify the best-performing product.
- To analyze sales by product and region.
- To generate meaningful insights from the dataset.
- To create a clean and structured analysis report.

3. Dataset Description

The project uses the sales dataset provided for the internship.

The dataset contains:

- Rows: 100
- Columns: 7

Columns

Column	Description
Date	Date of the sale
Product	Product sold
Quantity	Quantity sold
Price	Price of the product

| Customer_ID | Unique customer identifier |

| Region | Sales region |

| Total_Sales | Total sales amount |

The dataset contains sales records for different products, customers, and regions.

4. Technologies Used

- Python 3.x
- Pandas
- Visual Studio Code
- Git
- GitHub

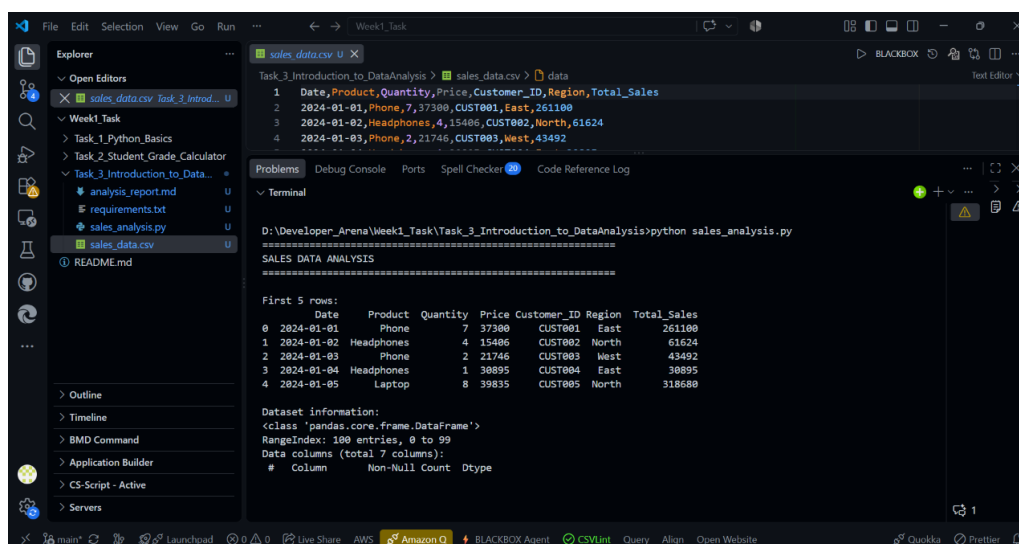
The main Python library used for the analysis is Pandas.

5. Project Structure

Task_3_Introduction_to_DataAnalysis/

- └─ sales_analysis.py
- └─ sales_data.csv
- └─ analysis_report.md
- └─ requirements.txt

##6. Visualization



The screenshot shows the Visual Studio Code interface with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project structure with files like `analysis_report.md`, `requirements.txt`, `sales_analysis.py`, and `sales_data.csv`. The code editor displays the `sales_data.csv` file with the following content:

```
1 Date,Product,Quantity,Price,Customer_ID,Region,Total_Sales
2 2024-01-01,Phone,7,37300,CUST001,East,261100
3 2024-01-02,Headphones,4,15400,CUST002,North,61624
4 2024-01-03,Phone,2,21745,CUST003,West,43492
```

The terminal shows the output of running `python sales_analysis.py`:

```
D:\Developer_Arena\Week1_Task\Task_3_Introduction_to_DataAnalysis>python sales_analysis.py

SALES DATA ANALYSIS
=====

First 5 rows:
   Date      Product  Quantity  Price Customer_ID Region  Total_Sales
0 2024-01-01   Phone         7  37300  CUST001   East    261100
1 2024-01-02 Headphones     4  15400  CUST002  North     61624
2 2024-01-03   Phone         2  21745  CUST003   West     43492
3 2024-01-04 Headphones     1  30895  CUST004   East     30895
4 2024-01-05  Laptop         8  39835  CUST005  North    318680

Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
...
```

Visual Studio Code interface showing a CSV file named `sales_data.csv` loaded into a Jupyter Notebook cell. The file explorer on the left shows the project structure, including `sales_data.csv` and `requirements.txt`. The Jupyter cell displays the first four rows of the CSV data:

```
1 Date,Product,Quantity,Price,Customer_ID,Region,Total_Sales
2 2024-01-01,Phone,7,37300,CUST001,East,261100
3 2024-01-02,Headphones,4,15406,CUST002,North,61624
4 2024-01-03,Phone,2,21746,CUST003,West,43492
```

The terminal output shows the dataset information:

```
Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 7 columns):
# Column Non-Null Count Dtype
---
0 Date 100 non-null object
1 Product 100 non-null object
2 Quantity 100 non-null int64
3 Price 100 non-null int64
4 Customer_ID 100 non-null object
5 Region 100 non-null object
6 Total_Sales 100 non-null int64
dtypes: int64(3), object(4)
memory usage: 5.6+ KB

Shape: (100, 7)
Columns: ['Date', 'Product', 'Quantity', 'Price', 'Customer_ID', 'Region', 'Total_Sales']
```

Visual Studio Code interface showing the same Jupyter Notebook cell, but now displaying the data types for each column:

```
Data types:
Date object
Product object
Quantity int64
Price int64
Customer_ID object
Region object
Total_Sales int64
dtype: object
```

The terminal output shows the basic statistics for the dataset:

```
Basic statistics:
      Quantity      Price      Total_Sales
count 100.000000  100.000000  100.000000
mean   4.780000  25808.510000 123650.480000
std    2.588163  13917.630242 100161.085275
min    1.000000   1308.000000   6540.000000
25%    2.750000  14965.250000  39517.500000
50%    5.000000  24192.000000  97955.500000
75%    7.000000  38682.250000 175792.500000
```

Visual Studio Code interface showing the same Jupyter Notebook cell, but now displaying the missing values by column:

```
Missing values by column:
Date      0
Product   0
Quantity  0
Price     0
Customer_ID 0
Region    0
Total_Sales 0
```

```
Task_3_Introduction_to_DataAnalysis > sales_data.csv > data
1 Date,Product,Quantity,Price,Customer_ID,Region,Total_Sales
2 2024-01-01,Phone,7,37300,CUST001,East,261100
3 2024-01-02,Headphones,4,15406,CUST002,North,61624
4 2024-01-03,Phone,2,21746,CUST003,West,43492
```

Missing values by column:

Date	0
Product	0
Quantity	0
Price	0
Customer_ID	0
Region	0
Total_Sales	0

dtype: int64
Total missing values: 0

Duplicate rows: 0
No missing values required cleaning.
No duplicate rows required cleaning.

=====

SALES ANALYSIS REPORT

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```
Task_3_Introduction_to_DataAnalysis > sales_data.csv > data
1 Date,Product,Quantity,Price,Customer_ID,Region,Total_Sales
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4 2024-01-03,Phone,2,21746,CUST003,West,43492
```

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SALES ANALYSIS REPORT

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Total sales: 12,365,048.00
Average sale: 123,650.48
Maximum sale: 373,932.00
Minimum sale: 6,540.00
Total quantity sold: 478
Unique products: 5
Unique customers: 100
Best-selling product by total sales: Laptop

Sales by product:

Product	Total Sales
Laptop	38892110
Tablet	2884340
Phone	2859394
Headphones	1384033
Monitor	1348071

```
Task_3_Introduction_to_DataAnalysis > sales_data.csv > data
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```

Unique customers: 100
Best-selling product by total sales: Laptop

Sales by product:

Product	Total Sales
Laptop	38892110
Tablet	2884340
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Headphones	1384033
Monitor	1348071

Sales by region:

Region	Total Sales
North	3983635
South	3737852
East	2519639
West	2123922

D:\Developer_Arena\Week1_Task\Task_3_Introduction_to_DataAnalysis>